



Department of Computer Science and Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: V Semester B.E Computer Science and Engineering

Course Code & Title: CB3491-Cryptography and Cyber Security

Name of the Faculty member (s): Mrs. K. Jeyakarthika, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV / Schnorr Digital Signature Scheme
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 13
- **Activity Chosen:** Think-Pair-Share
- **Justification:**
 - Think pair share activity involves the students to think individually and to share the knowledge with classmates.
 - In Digital Signature Scheme signature creation and verification process is a tedious process.
 - Through this activity, students can able to create and verify the digital signature and then discuss their observations with a partner.
 - This enables them to apply their knowledge, identify specific scenarios and understand how to solve them efficiently.
- **Time Allotted for the Activity:** 30 minutes
- **Details of the Implementation**
 - Think-Pair-Share innovative practice conducted for III year CSE students, after explanation of the concept Schnorr Digital Signature Scheme.
 - The instructor explained the problems in the concept of Schnorr Digital Signature Scheme to the students. 3 questions were given to students and asked them to generate a key and signature.
 - Students were given 15 minutes to independently think of their understanding of the given problem of how the key generation, signature generation and verification will happen using digital signature concept.
 - Following this, the instructor encouraged students to pair up with nearby classmates to discuss their solutions shown in Figure 1. Students then shared their solutions with the entire class during the last segment of 15 minutes.
 - The instructor insisted one student to present, thereby Anna Lakshmi of III CSE has

solved the given problem shown in Figure 2.

- The instructor moderated the discussion, emphasizing key points.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO12	PSO1	PSO2	PSO3
CO4	3	3	3	3	3	1	2	1	3	3	2

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5
	3	3	3	3	3
Justification for correlation	Apply the basic Computer Science Engineering and mathematics Knowledge for performing Symmetric encryption.	Analyzing various classification of cybercrimes are highly correlated	Developing systems with basic cybercrime mechanisms are highly correlated.	Investigate various tools of cybercrimes is highly correlated.	Use the virtual environment and crypt tool for performing digital signature.
	PO8	PO9	PO12		
	1	2	1		
	To complete the assignments/ tasks in an ethical way.	The students will able to work as an individual and team to how to distribute a key with integrity and authentication.	Updating the digital signature algorithms to the current trends are moderately correlated.		

• **Images / Screenshot of the practice:**



Figure 1 – Students Think and Share their Thoughts



Figure 2 – Students Think and Share their Thoughts and Topic Presentation

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Working as a pair, they can share their knowledge with them and come up with better understanding about the topic.
 - Different types of problems and complex solutions are discussed in the class room, so students come to know how to solve the mathematical area.

- ❖ ***Benefit of the practice:***

- Students are actively engaged.
 - Students learn from each other.
 - Builds a friendly, yet academic atmosphere.
 - Includes all the students in the teaching-learning process.

- ❖ ***Challenges faced in implementation:***

- Students felt very difficult in this activity and could not complete in time whenever some difficult question is given.
 - Students are hesitating to speak in front of all.
 - It took a longer time than other strategies.

References:

- ❖ <http://www.et.iitb.ac.in/TeachingResources.html>
- ❖ <https://www.readingrockets.org/strategies/think-pair-share>
- ❖ <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improveyour-classroom-discussions1704.html>
- ❖ <https://serc.carleton.edu/introgeo/interactive/tpshare.html>

Signature of Faculty Member

HOD-CSE